

7) Use De Morgan's laws to find the negation of each of the following statements

a) Jan is rich and happy.

Soln: p : Jan is rich q : Jan is happy

$$\neg(p \wedge q) \equiv \neg p \vee \neg q \text{ (1st De Morgan's Law)}$$

Jan is not rich or Jan is not happy. (English representation)

b) Carlos will bicycle or run tomorrow.

Soln: p : Carlos will bicycle tomorrow q : Carlos will run tomorrow

$$\neg(p \vee q) \equiv \neg p \wedge \neg q \text{ (2nd De Morgan's Law)}$$

Neither Carlos will bicycle nor run tomorrow (English representation)

c) Mei walks or takes the bus to class.

Soln: p : Mei walks to class q : Mei takes the bus to class.

$$\neg(p \vee q) \equiv \neg p \wedge \neg q \text{ (2nd De Morgan's Law)}$$

Neither Mei walks nor takes the bus to class (English representation)

d) Ibrahim is smart and hardworking.

Soln: p : Ibrahim is smart q : Ibrahim is hardworking

$$\neg(p \wedge q) \equiv \neg p \vee \neg q \text{ (1st De Morgan's Law)}$$

Ibrahim is not smart or Ibrahim is not hardworking.

9) For each of these compound propositions, use the conditional-disjunction equivalence (Example 3) to find an equivalent compound proposition that does not involve conditionals.

a) $p \rightarrow \neg q \equiv \neg p \vee \neg q$ (By conditional-disjunction equivalence that $p \rightarrow q \equiv \neg p \vee q$)

b) $(p \rightarrow q) \rightarrow r \equiv (\neg p \vee q) \rightarrow r$ (By conditional-disjunction equivalence)

$$\equiv \neg(\neg p \vee q) \vee r \text{ (By conditional-disjunction equivalence)}$$

$$\equiv (\neg(\neg p) \wedge \neg q) \vee r \text{ (By 2nd De Morgan's Law)}$$

$$\equiv (p \wedge \neg q) \vee r \text{ (Double negation law)}$$

$$\equiv (p \wedge \neg q) \vee r \vee (p \wedge \neg q) \text{ (1st commutative law)}$$